

CODE SPORT



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In this month's column, we feature a set of computer science interview questions.

Over the last couple of months, we have been discussing a set of questions on machine learning and data management. This month, we return to our discussion of topics in natural language processing (NLP). We focus on the important problem in NLP known as parts of speech (POS) tagging, covering why it is important and some of the techniques used in this area.

Parts of speech tags are used to tag words in a sentence with their syntactic category, such as whether a word is a noun, verb, adjective or adverb. For instance, given the sentence, 'The dog barks,' its POS tags would be as follows: 'The (determiner) dog (noun) barks (verb).' The tags in brackets indicate the parts of speech associated with the word in the sentence. Some readers may wonder why one needs to do POS tagging. Isn't the POS tag of a word known automatically? Well, the answer is not so straightforward.

Most of the words in the English language (and in the case of most other languages as well) can take different tags based on the context in which they are used. For example, consider the two sentences:

- (1) Please give me the book.
- (2) Please book the evening flight to Delhi.

In Sentence 1, the word 'book' is associated with the noun POS tag. In Sentence 2, the same word is associated with the verb POS tag. Certain tags such as prepositions in English are easier to tag in a sentence, since they are associated with an unambiguous POS tag. On the other hand, most other words have an ambiguous association with POS tags, since they can take different tags based on the context. POS tags can be divided into two broad categories, namely, Closed Class Tags and Open Class Tags. In the first category, member words are typically fixed and form a closed set. Open Class Tags are those whose membership can increase

over time due to the addition of new words. For example, a preposition is a Closed Class Tag since, typically, in English, new prepositional words don't get added. On the other hand, consider tag classes such as nouns and verbs. In English, new nouns can get added due to new entities such as 'Google' or 'iPhone' and new verbs can get added, such as 'fax', etc. Hence, verbs and nouns are Open Class Tags.

Apart from POS tag classes such as nouns, verbs, prepositions, etc, how many POS tag classes are typically considered for POS tagging of English sentences? While there are different tag sets available with differing number of tags, a popular one in the NLP community is the Penn Tree Bank tag set of 45 tags, available at: https://www.ling.upenn.edu/courses/Fall_2003/ling001/penn_treebank_pos.html

To those still wondering why POS tagging is needed for text, it is a pre-processing step that's necessary in a number of related problems, namely, information retrieval (IR), text-to-speech processing, word sense disambiguation, etc. POS tagging is also useful for syntactic parsing, since the tag associated with a word can be used to infer syntactic classes of surrounding words. For instance, nouns are preceded by determiners and adjectives, and verbs are typically followed by nouns, and so on. There has been considerable prior work in using POS tagging to improve IR performance, by cutting down the terms indexed based on their POS tags. For instance, since indexing only the nouns in a document generally provides a good idea on what the document is about, it is possible to create a pruned index set using POS tags. Similarly, POS tags can be used to improve text-to-speech systems, in deciding how a word should be spoken with different intonations. For instance, consider the following two sentences: (1) I object to your insolent tone.